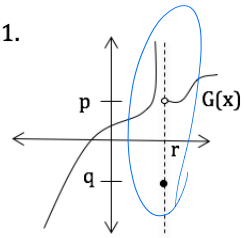
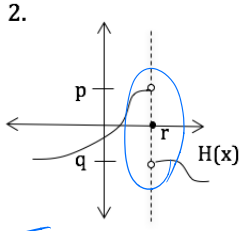
Math 161: Continuity at a Point

1.



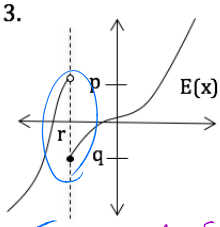
Asymptote, Jump $\lim_{x \to r^-} G(x)$ DNE

2.



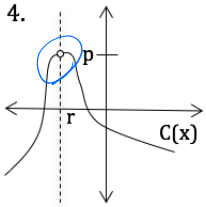
Jump $\lim_{x \to r} H(x)$ DNE

3.



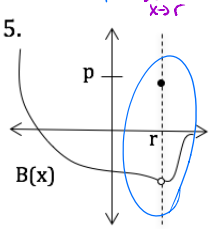
Jump $\lim_{x \to r} E(x)$ DNE

4.



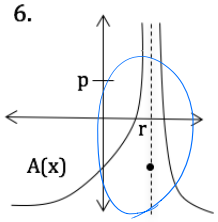
Hole C(r) is undefined

5.



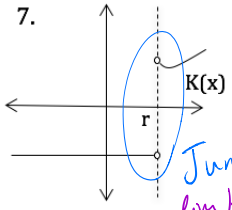
Jump $\lim_{x \to r} B(x) \neq B(r)$

6.



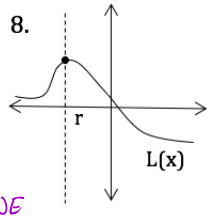
Asymptote $\lim_{x \to r} A(x)$ DNE

7.



Jump $\lim_{x \to r} K(x)$ DNE

8.



Continuous!

$\lim_{x \to r} L(x) = L(r)$

1. For each of the above graphs, state at which points (if any) the function appears to be discontinuous. In your own words, describe why you think there is a discontinuity at that point.

2. Now we introduce the formal definition of continuity at a point. State the three criteria for continuity at a ponit: $f(x)$ is continuous at $x = c$ if:

1) $\lim_{x \to c} f(x)$ exists

2) $f(c)$ is defined

3) $\lim_{x \to c} f(x) = f(c)$

Using the formal definition, go back and explain precisely why the functions are discontinuous (Which of the three criteria are not satisfied?).